Bardiya Kariminia

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Education

Shahid Beheshti University, BS in Computer Engineering — Sep 2022 – Ongoing

- Overall GPA: **18.78/20.00**

Teaching Assistantships:

- Machine Learning, Artificial Intelligence, Linear Algebra, Advanced Programming, Discrete Math and Statistics, Computer Architecture, Operating Systems.

Research Interests

- **Generative AI**: Diffusion Models, Diffusion Transformers, GANs, Variational Autoencoders
- **Vision-Language Models**: Multimodal learning, Hallucination detection and pattern analysis
- **Neural Rendering**: NeRF, Gaussian Splatting, 3D Reconstruction
- **Optimization**: Bayesian Optimization, Evolutionary Algorithms

Publications

A Race Bias Free Face Aging Model for Reliable Kinship Verification — **Link**

Ali Nazari, **Bardiya Kariminia**, Mohsen Ebrahimi Moghaddam.

arXiv preprint, 2025.

UHP Detection: LVLMs have their Unique Hallucination Pattern in the Consistency Space — **Link**

Amir Mohammad Ezzati, Kiyan Rezaee, **Bardiya Kariminia**, Mohamad Amin Yousefi, Asal Mohammadjafari Mamaqani, Behrad Samimi, Mohammad Hossein Rohban.

arXiv preprint, 2026.

SIA: Selective Image Analogy — **Link**

Amirhossein Alimohammadi, Kian Izadpanah, **Bardiya Kariminia**, Ali Mahdavi Amiri.

SIGGRAPH Asia 2026.

Experience

Volunteer BCs student at GruviLab, Simon Fraser University. — Apr 2026 – Ongoing

- Developed SIA, a framework for solving the Image Analogy task in image editing upon *FLUX.2 Klein* diffusion transformer, introducing a Selective LoRA and a Constraint Support Vector module to help the model suppress unintended edits more efficiently. **Accepted on SIGGRAPH Asia 2026**
- Proposed a new problem formulation, Multi-View Image Analogy, extending image analogy to jointly transfer camera information under uncalibrated camera settings.
- Developed GAIA (Geometry-Aware Image Analogy), a framework for solving Multi-View Image Analogy by modifying RoPE positional encoding and its offset computation, leveraging a dense feature matching backbone for offset estimation. **Submitted to ICLR 2027.**
- Created the GAIA dataset for Multi-View Image Analogy, comprising over 100k samples curated for this newly proposed problem statement.

Research Internship at AIDAM Group; Max Planck Institute of Informatics; supervisors: Prof. Vahid Babaei. — May 2025 - Mar 2026

The research internship mainly focused on artificial intelligence-aided engineering designs using Diffusion models. Currently, **manuscript in preparation**.

- Developed a new training technique to add diversity via Determinental Point process to Diffusion model in the forwarding phase for engineering designs such as structural 2D beams and wind turbines airfoil.

- optimized designs Pareto-front performance problem using the system's diversity injection ability via Bayesian Optimization and Evolutionary Algorithm such as *NSGAII*.
- Gained in-depth experience with optimization techniques and diversity prepending.
- techniques: DDIM, DDPM, Bayesian Optimization, Multi-Objective Determinantal point process, Adversarial training.

Research Assistant at RIML Group; Sharif University of Technology; supervisors: Prof. Mohammad Hossein Rohban. Aug 2025 - Mar 2026

- Investigated the consistency space of Vision-Language Models (VLMs) across image and text modalities, analyzing logical polarity to uncover hallucination patterns in black-box settings.
- Designed a framework to detect and learn model-specific hallucination patterns using the extracted information from logical polarity and their feature, aiming to identify unique behavioral signatures for each VLM architecture.
- techniques: LVLM, Hallucination detection, pattern matching.

Research Internship, Shahid Beheshti Computer Vision Lab; supervisor: Prof. Mohsen Ebrahimi Moghadam. Sep 2024 - July 2025

Developed a deep generative framework for face age Transformation using generative Models used to improve Kinship accuracies.

- Developed a facial image transformation system capable of aging and de-aging faces while maintaining identity, ethnicity, and high visual realism. The work introduced a novel approach addressing racial fairness in image generation, achieving promising improvements in kinship verification accuracy—ranging from 0.39% up to 5.22%.
- techniques: GANs, feature extraction, data balancing, cyclic domain learning.

Projects

Age Transformation without racial bias for kinship verification **Github Link**

- The main project for the Shahid Beheshti University computer vision internship focused on transforming face images across different ages in a bias-free manner.

Diff-MOBO **Github Link**

- The main project for the Max Planck research internship.

UHP Detection: LVLM Hallucination Pattern Analysis **Github Link**

- Main project for the UHP Detection paper, implementing a framework to detect and analyze hallucination patterns in Vision-Language Models using consistency space and logical polarity.

Skills

- Python, NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn
- **Image processing Libraries**: OpenCV, Pillow
- **AI frameworks**: JAX, Transformers, Diffusers, gsplat, DeepFace, PyTorch, TensorFlow, TorchVision
- LaTeX
- **English**: Fluent